



2201 321 Veterinary Virology



Family *Rhabdoviridae* and Family *Birnaviridae*

Wilasinee Surasa, DVM, M.Sc.

Lecturer, Faculty of Veterinary of Sciences

Mahasarakham University

วัตถุประสงค์

- นิสิตมีความรู้ความเข้าใจเกี่ยวกับไวรัสและโรคที่เกิดจากไวรัสใน Family *Rhabdoviridae* และ Family *Birnaviridae* ดังนี้
 - ลักษณะและชีววิทยาของไวรัส
 - การก่อโรคในสัตว์ พยาธิกำเนิดและอาการแสดงทางคลินิก
 - ตัวอย่างเพื่อชั้นสูตรทางห้องปฏิบัติการ วิธีการ และเทคนิคที่ใช้ชั้นสูตรทางห้องปฏิบัติการ
 - การควบคุมและการป้องกันโรค

Family *Rhabdoviridae*



- The family *Rhabdoviridae* includes 56 genera, 434 species that are assigned to four subfamilies;
 - Subfamily *Alpharhabdovirinae*
 - Includes 32 genera for viruses infecting only vertebrates, only invertebrates or vertebrate hosts and arthropod vectors
 - Subfamily *Betarhabdovirinae*
 - Includes six genera for viruses infecting plant hosts and arthropod vectors
 - Subfamily *Gammarhabdovirinae*
 - Includes only a single genus for viruses infecting finfish
 - Genera not assigned to a subfamily
 - Within the *Rhabdoviridae*, seven genera do not fall within any of the three existing subfamilies

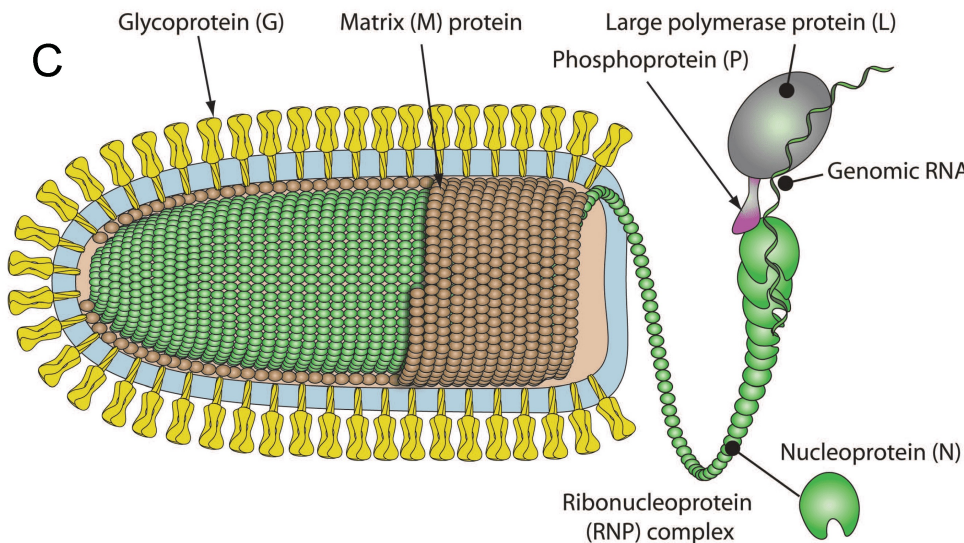
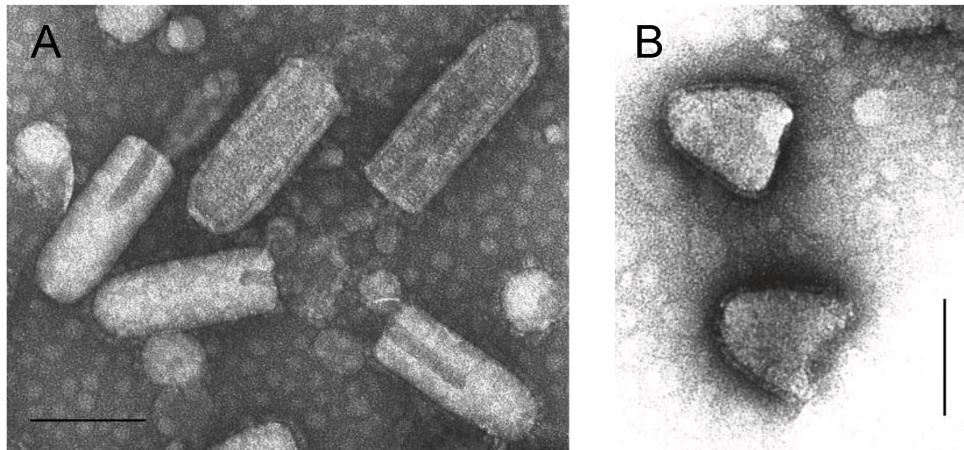
Viral properties

- Bullet-shaped or cone-shaped (rhabdo = rod)
- Size: 100–460 nm in length and 45–100 nm in diameter
- Enveloped with large spikes (peplomers)
 - which are 5–10 nm long and about 3 nm in diameter
- The nucleocapsid (30–70 nm in diameter) has helical symmetry
- Genome
 - A single molecule of linear, negative-sense single-stranded RNA
 - Size: approximately 10–16 kb
- Virus infectivity is rapidly inactivated at 56°C, or following UV-, gamma- or X-irradiation, or exposure to formalin or lipid solvents such as detergents



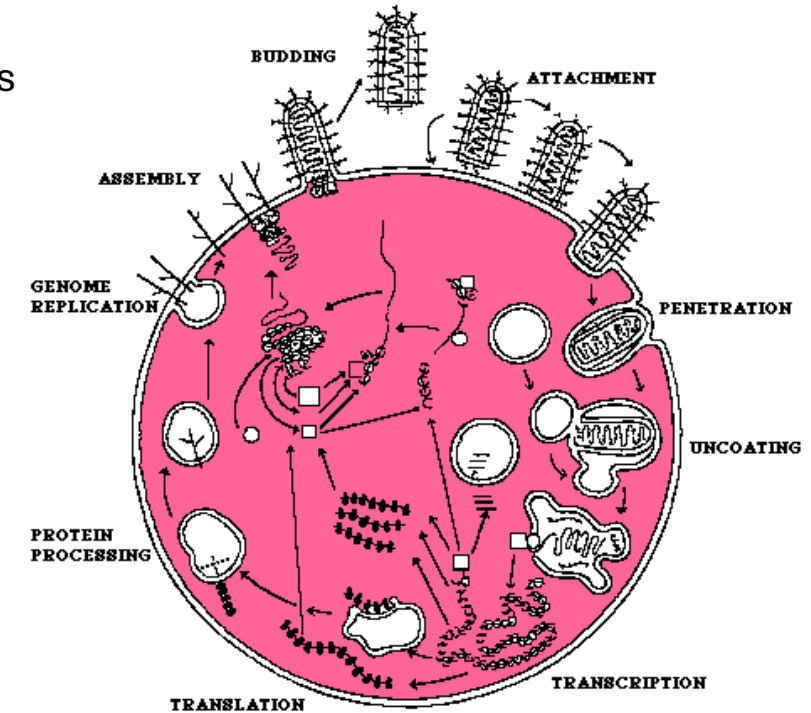
Structural proteins

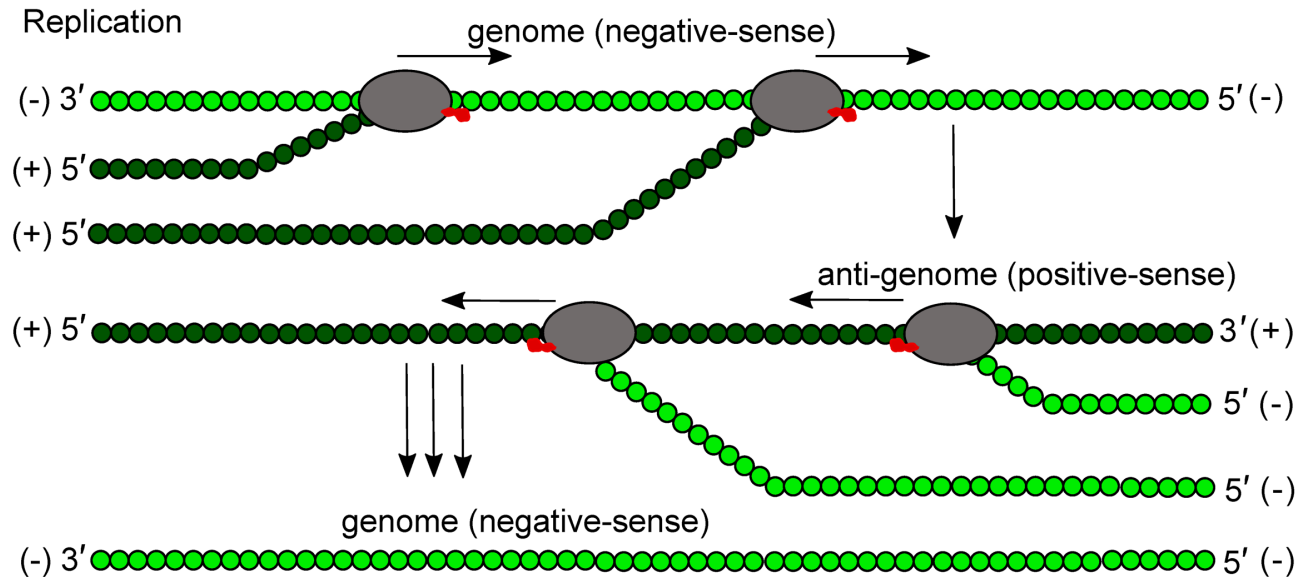
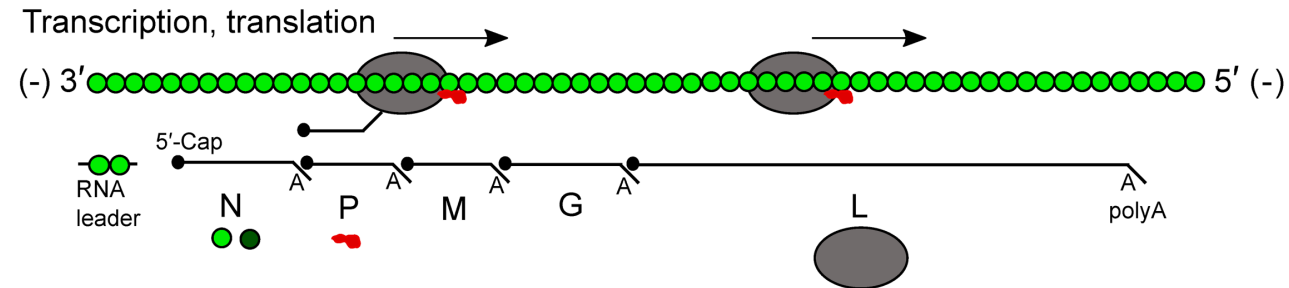
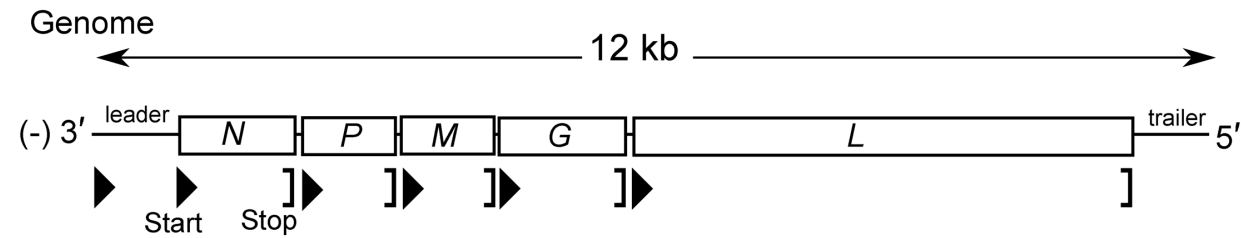
- The nucleocapsid consists of a ribonucleoprotein (RNP) complex comprising the genomic RNA and tightly bound nucleoprotein (N)
 - RNA polymerase (L) and polymerase-associated phosphoprotein (P)
- mRNAs template; translate into 5 proteins:
 - N = nucleoprotein (nucleocapsid)
 - L = RNA-dependent RNA polymerase (transcriptase)
 - P = phosphoprotein mediates binding of L protein to nucleocapsid (a cofactor of the viral polymerase)
 - G = glycoprotein binds to host cell receptor (envelope spikes)
 - M = binds to nucleocapsids and the cytoplasmic domain of G



Virus replication

- From studies of vesiculoviruses and lyssaviruses
 - Cytoplasmic replication
 - Leader RNA of some viruses has been identified in the nucleus of infected cells
- Rhabdovirus adsorption is mediated by G attachment to cell surface receptors, and penetration of the cell occurs by endocytosis via coated pits
- Low pH within the endosome triggers fusion between endosomal and viral membranes
- Once the nucleocapsid is released into the cytoplasm, the RNA genome is repetitively transcribed (primary transcription) by the virion transcriptase
- Following translation, RNA replication occurs in the cytoplasm (full-length positive-sense and then full-length negative-sense RNA synthesis)
- G proteins (from endoplasmic reticulum membrane, and/or Golgi apparatus) move to the plasma membrane
- Maturation: Viral nucleocapsid structures are assembled in association with M and lipid envelopes containing viral G to form virions -> budding through plasma membrane





Rhabdoviruses affected in animals



- Subfamily *Alpharhabdovirinae*
 - Genus *Ephemerovirus*
 - Bovine Ephemeral Fever Virus
 - Genus *Lyssavirus*
 - Rabies Virus
 - Genus *Vesiculovirus*
 - Vesicular Stomatitis Virus

Bovine Ephemeral Fever (BEF)

- 3-day stiff-sickness: ไช้สามวัน, ไช้ขาแข็ง
- Arthropod-transmitted disease: *Culicoides* species, anopheline and culicine mosquito species
 - Transmission by contact or fomites does not occur
- Hosts: cattle and water buffalo
- Distribution
 - Tropical and subtropical zones of Africa, Australia, Middle East, **Asia**
 - No report in North America and Europe
- Morbidity rates may be as high as 80%; overall mortality rate is usually 1%–2%, although it can be higher in lactating cows, bulls in good condition, and fat steers (10%–30%)



Pathogenesis

BPF is most prevalent in the wet season in the tropics and in summer to early autumn in the subtropics or temperate regions (when conditions favor multiplication of biting insects)

Macroscopic lesions

- Serofibrinous polyserositis
- Synovitis
- Pulmonary and lymph node edema
- Focal necrosis of muscles

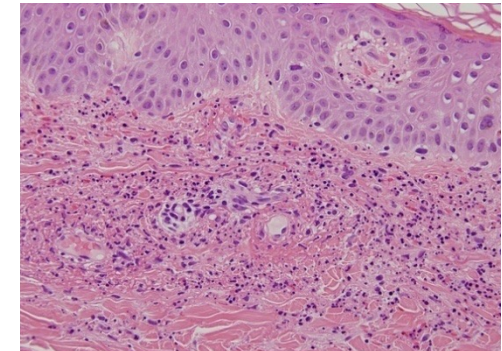
Transmitted by IV inoculation

Lymphoid tissue

Infection results in longterm immunity

Viremia

Microscopic lesions -
Perivascular neutrophilic infiltration



Affected cattle may become recumbent for 8 hours to > 1 week

Spontaneous recovery usually occurs within a few days

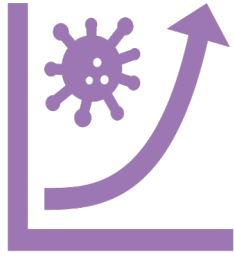


Clinical signs

- Clinical signs occur suddenly and vary in severity, can include the following:
 - Biphasic to polyphasic fever (40° – 42°C [104° – 107.6°F])
 - Shivering
 - Anorexia, inappetence and drooling
 - Tearing and/or serous nasal discharge
 - Stiffness and lameness
 - Sudden decrease in milk yield
 - Increased heart rate
 - Pulmonary emphysema
 - Tachypnea or dyspnea
 - Diarrhea



Diagnosis and control



Diagnosis

- Clinical signs
- Virus isolation; mosquito (*Aedes albopictus*) cells or suckling mouse brain
- RT-PCR assays
- ELISA
- Immunofluorescence
- Agar gel precipitin tests



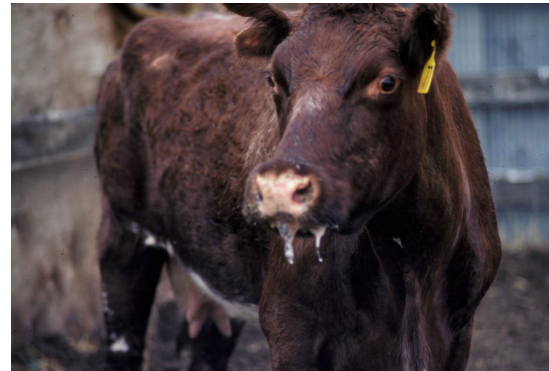
Control

- Vector control*



Rabies Virus

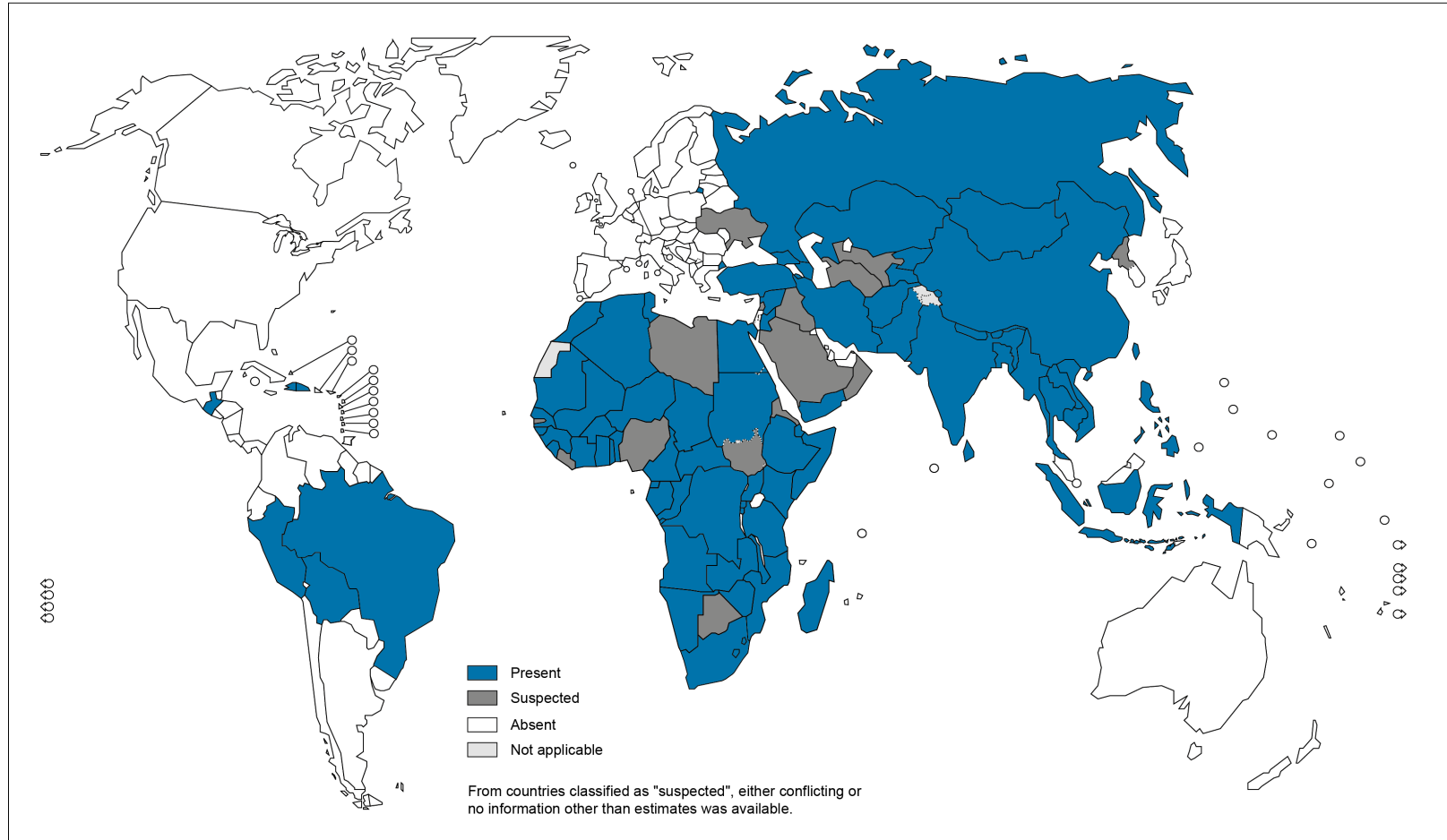
- Hosts
 - All mammals**
- Reservoir hosts: dog, fox, wolf, jackal, raccoon, skunk, mongoose and bats
- Transmissions
 - Bite of rabid animals
 - Corneal transplantation in human
 - Aerosol*



Family Rhabdoviridae and Family Birnaviridae

Rabies, countries or areas at risk

Presence of dog-transmitted human rabies based on most recent data points from different sources, 2010-2014

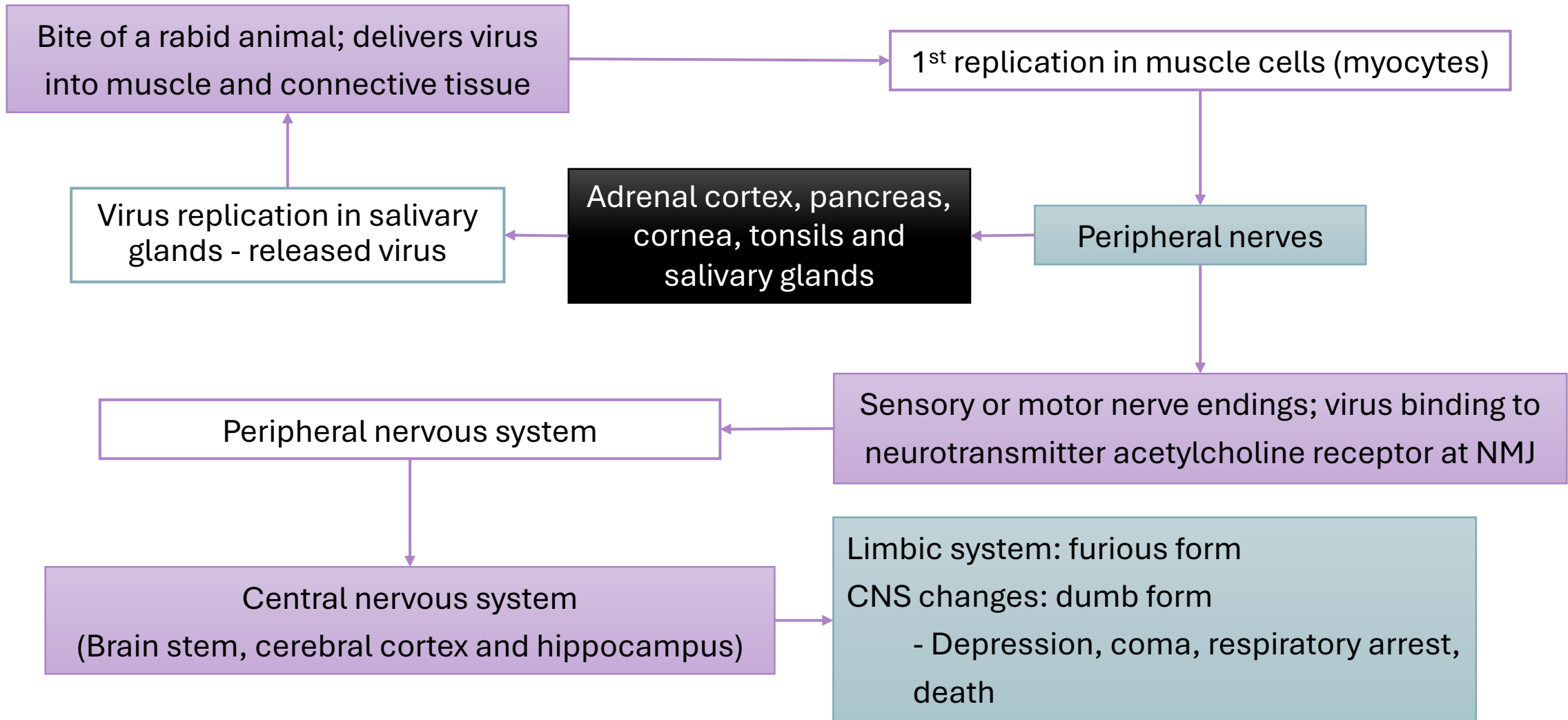


The boundaries and names shown and the designations used on this map do not imply the expression of any opinion whatsoever on the part of the World Health Organization concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted lines on maps represent approximate border lines for which there may not yet be full agreement. © WHO 2015. All rights reserved

Data Source: World Health Organization
 Map Production: Control of Neglected
 Tropical Diseases (NTD)
 World Health Organization



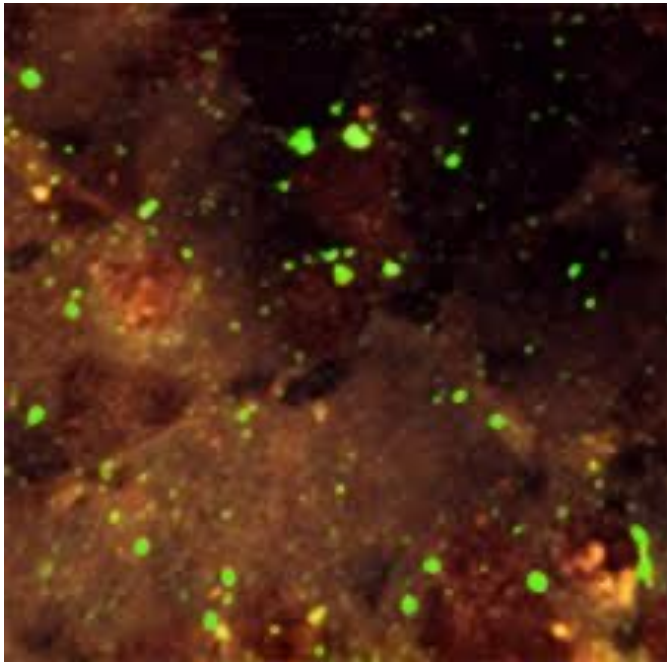
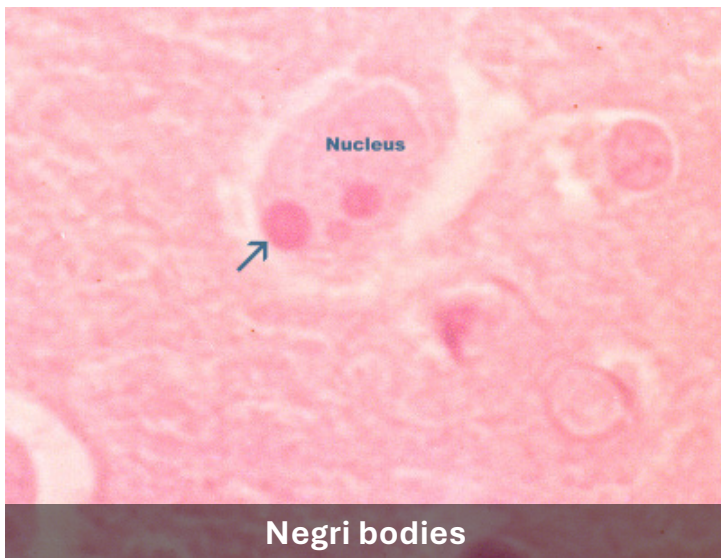
Pathogenesis



Clinical signs

- Similar in most species but variation between individuals
- Incubation period: 14 - 90 days (or longer)
- Furious form
 - Nervous, aggressive, bites without provocation, pharyngeal paralysis - cannot swallow water 'hydrophobia'
 - Other signs: excessive salivation, exaggerated responses to light and sound, hyperesthesia (bite and scratch themselves)
- Dumb (or paralytic) form
 - Higher mental function
 - Paralysis of throat, masseter muscles, profuse salivation, inability to swallow, dropping of the lower jaw
- Terminally
 - Convulsive seizures
 - Coma
 - Respiratory arrest
 - Death
 - 2-14 days after onset of clinical signs





Diagnosis

- Clinical signs: highly variable
- Negri bodies: difficult to identify
 - Suspect animal must be killed, and brain tissue collected for testing
- Mouse inoculation test
- Rabies virus antigen
 - Brain tissue: medulla, cerebellum and hippocampus
 - Immunofluorescence
 - Immunohistochemical staining
- Viral RNA: RT-PCR assay

Rabies pre-exposure prophylaxis

- Rabies pre-exposure vaccination (PrEV); updates to the Advisory Committee on Immunization Practices (ACIP) recommendations to prevent human rabies, 2022
 - A 2-dose PrEV schedule has replaced the 3-dose PrEV schedule to protect people from rabies for up to 3 years. Options for maintaining protection beyond 3 years are also described.
 - Risk categories have been redefined into 5 risk groups.
 - The minimum acceptable laboratory value (antibody titer) used to determine whether rabies vaccine booster doses are needed was revised and standardized.
 - Many people for whom serial titers were recommended every 2 years now require only a one-time titer (and booster if below a certain level) or a one-time booster.



Clinical guidance for administering PrEV to people (From CDC)

Risk category	Who this typically affects	Recommendations
Risk category 1 Highest risk	People who work with live or concentrated rabies virus in laboratories	- 2 doses, days 0 and 7 - Check titer every 6 months
Risk category 2	People who frequently do at least one of the following: handle bats, have contact with bats, enter high-density bat environments like caves, or perform animal necropsies	- 2 doses, days 0 and 7 - Check titer every 2 years
Risk category 3	People who interact with, or are at higher risk to interact, with mammals other than bats that could be rabid, for a period longer than three years after they receive PrEP. This group includes: Most veterinarians, veterinary technicians, animal control officers, wildlife biologists, rehabilitators, trappers, and spelunkers (cave explorers) Certain travelers to regions outside of the United States where rabies in dogs is commonly found	- 2 doses, days 0 and 7, plus: - Either a one-time titer check after 1 year and up to 3 years following the first 2-dose vaccination Or - 1-dose booster between 3 weeks and 3 years following the first vaccine in the 2-dose vaccination
Risk category 4	Same population as risk category 3, but at a higher risk for ≤ three years after they receive PrEV	- 2 doses, days 0 and 7
Risk category 5 Lowest risk	General U.S. population	None

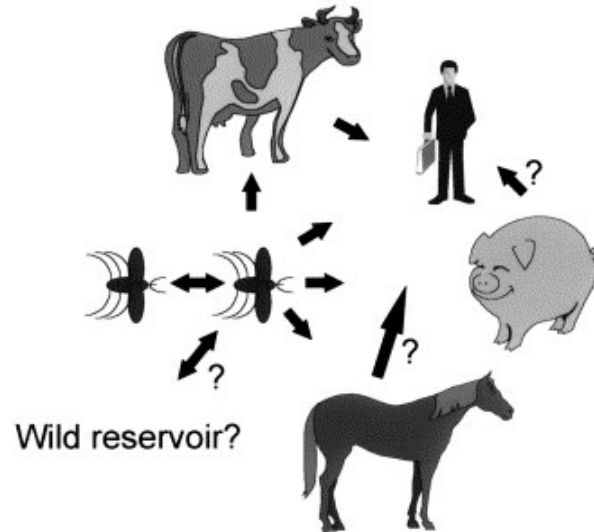
Thai guideline Rabies pre-exposure prophylaxis, 2018

ประชาชนทั่วไปที่ต้องการฉีดวัคซีนก่อนสัมผัสโรค		ผู้ที่มีปัจจัยเสี่ยงสูงในการสัมผัสโรคตลอดเวลาหรือผู้ที่มีภูมิคุ้มกันบกพร่อง
การฉีดเข้ากล้ามเนื้อ (intramuscular regimen: IM)	ใช้วัคซีนชนิด PVRV, CPRV, PCECV, PDEV 1 เข็ม (0.5 or 1 ml) ฉีดเข้ากล้ามเนื้อเนื้อต้นแขนในวันที่ 0 และ 7	ใช้วัคซีนชนิด PVRV, CPRV, PCECV, PDEV 1 เข็ม (0.5 or 1 ml) ฉีดเข้ากล้ามเนื้อเนื้อต้นแขนในวันที่ 0, 7 และ 21 หรือ 28
การฉีดเข้าในหนัง (intradermal regimen: ID)	ใช้วัคซีนชนิด PVRV (Verorab®), CPRV, PCECV 0.1 ml/จุด จำนวน 2 จุด ฉีดเข้าในผิวหนังบริเวณต้นแขน 2 ข้าง ในวันที่ 0 และ 7 หรือ 21	ใช้วัคซีนชนิด PVRV (Verorab®), CPRV, PCECV 0.1 ml/จุด จำนวน 1 จุด ฉีดเข้าในผิวหนังบริเวณต้นแขนในวันที่ 0, 7 และ 21 หรือ 28

(สถานเสาวภา, สภากาชาดไทย)

**เนื่องจากกลุ่มอาชีพ เช่น สัตวแพทย์, เจ้าหน้าที่ห้องปฏิบัติการที่ทำงานกับเชื้อไวรัสโรคพิษสุนัขบ้า ฯลฯ มีความเสี่ยงสูงในการติดเชื้อ จึงควรมีระดับภูมิคุ้มกันในการป้องกันโรคพิษสุนัขบ้าที่สูงอย่างต่อเนื่องตลอดเวลา เพื่อกรณีที่มีการสัมผัสโรคโดยไม่รู้ตัว สถานเสาวภาจึงยังแนะนำให้บุคคลกลุ่มนี้ได้รับการฉีดวัคซีนแบบก่อนสัมผัสโรค จำนวน 3 ครั้ง

Proposed natural cycle of VSV

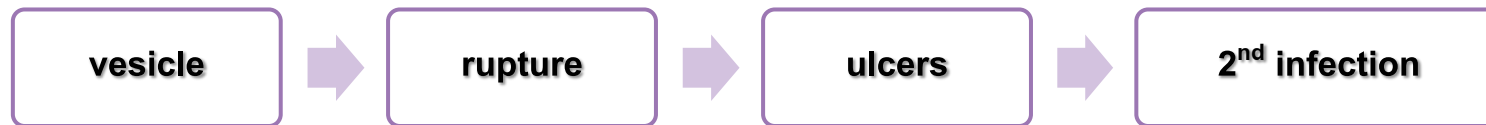


Vesicular Stomatitis

- Distribution: Americas
- Vesicular Stomatitis Virus
 - New Jersey: North America
 - Indiana: North America
 - Alagoas: Brazil
 - Cocal: Trinidad and Brazil
- Hosts: cattle, pigs, horses
- Vesicular lesions
 - Differential diagnosis from foot-and-mouth disease

Pathogenesis

- Transmission
 - Direct contact
 - Biting insects
 - Sandflies: transovarial transmission
- Incubation period: 1-5 days
- Lesion: vesicle



- Sites: tongue, oral mucosa, teats, coronary bands, snout

Clinical signs

- First signs of infection
 - Cattle and horses: excess salivation and fever
 - Swine:
 - Lameness
 - Heal within 7–10 days
 - Not persistence immunity, they can re-infect again**



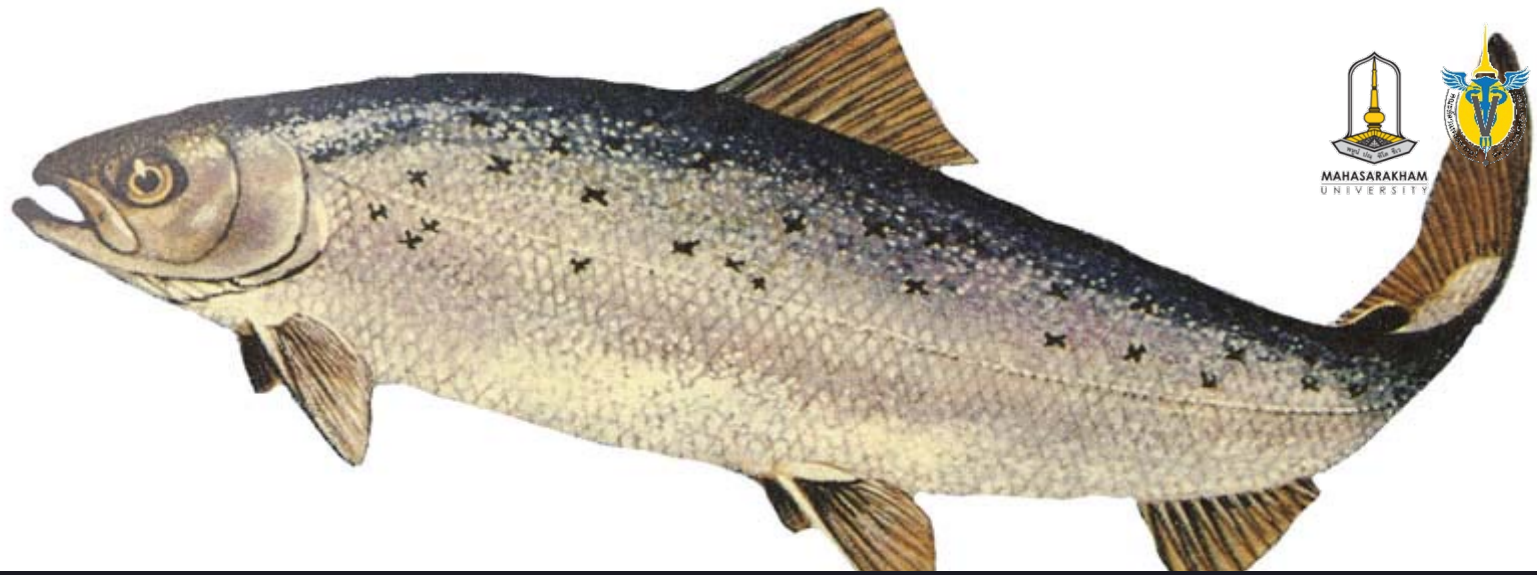
FIGURE 18.8 Extensive (craterous) ulcer in the tongue of a horse with vesicular stomatitis. (Courtesy of R. Bowen, Colorado State University.)

Laboratory diagnosis

- Specimens
 - Vesicles, vesicular fluid, saliva or serum
- Detect antigen
 - Electron microscopy
 - RT-PCR
 - Isolation into cell culture: CPE
- Detect antibody
 - ELISA



Family *Birnaviridae*

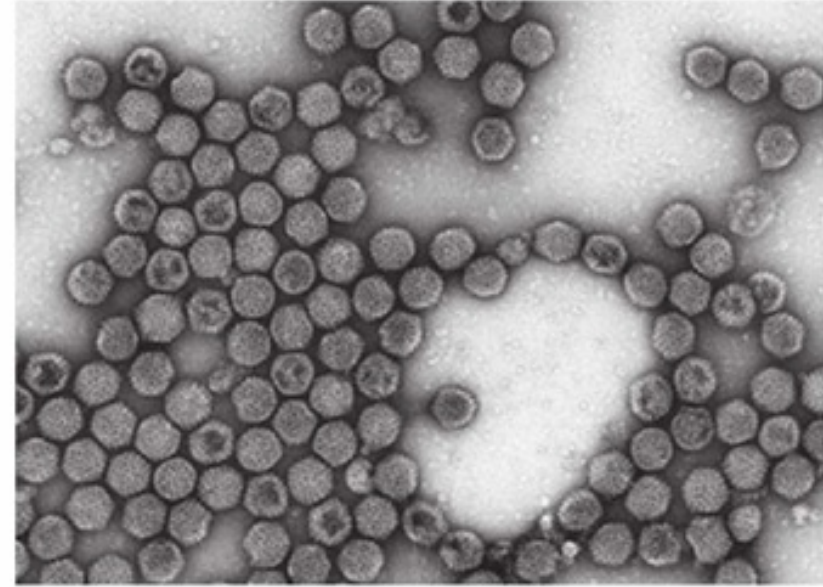


- Family *Birnaviridae* comprise 3 genera
 - Genus *Avibirnavirus*
 - Infectious bursal disease virus*
 - Genus *Aquabirnavirus*
 - Infectious pancreatic necrosis virus*
 - Viruses of mollusks and crustaceans
 - Genus *Entomobirnavirus*
 - Infect only insects

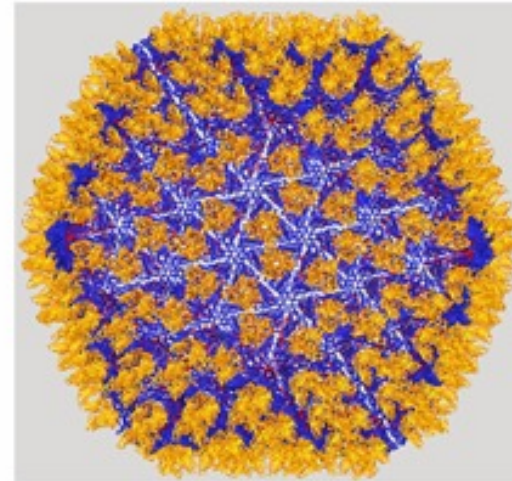
Viral Properties

- Non-enveloped
- Approximately 60 nm in diameter
- Two molecules of linear double-stranded RNA
- Hexagonal, Icosahedral symmetry
- Stable in environment
- Replication in cytoplasm
- Found cytopathic effect in cell culture

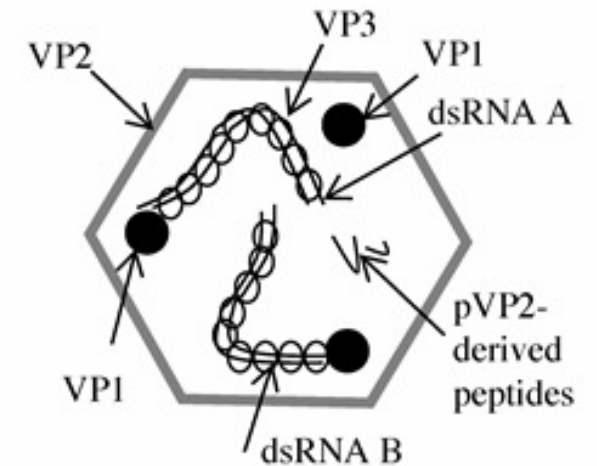
(A)



(B)



(C)



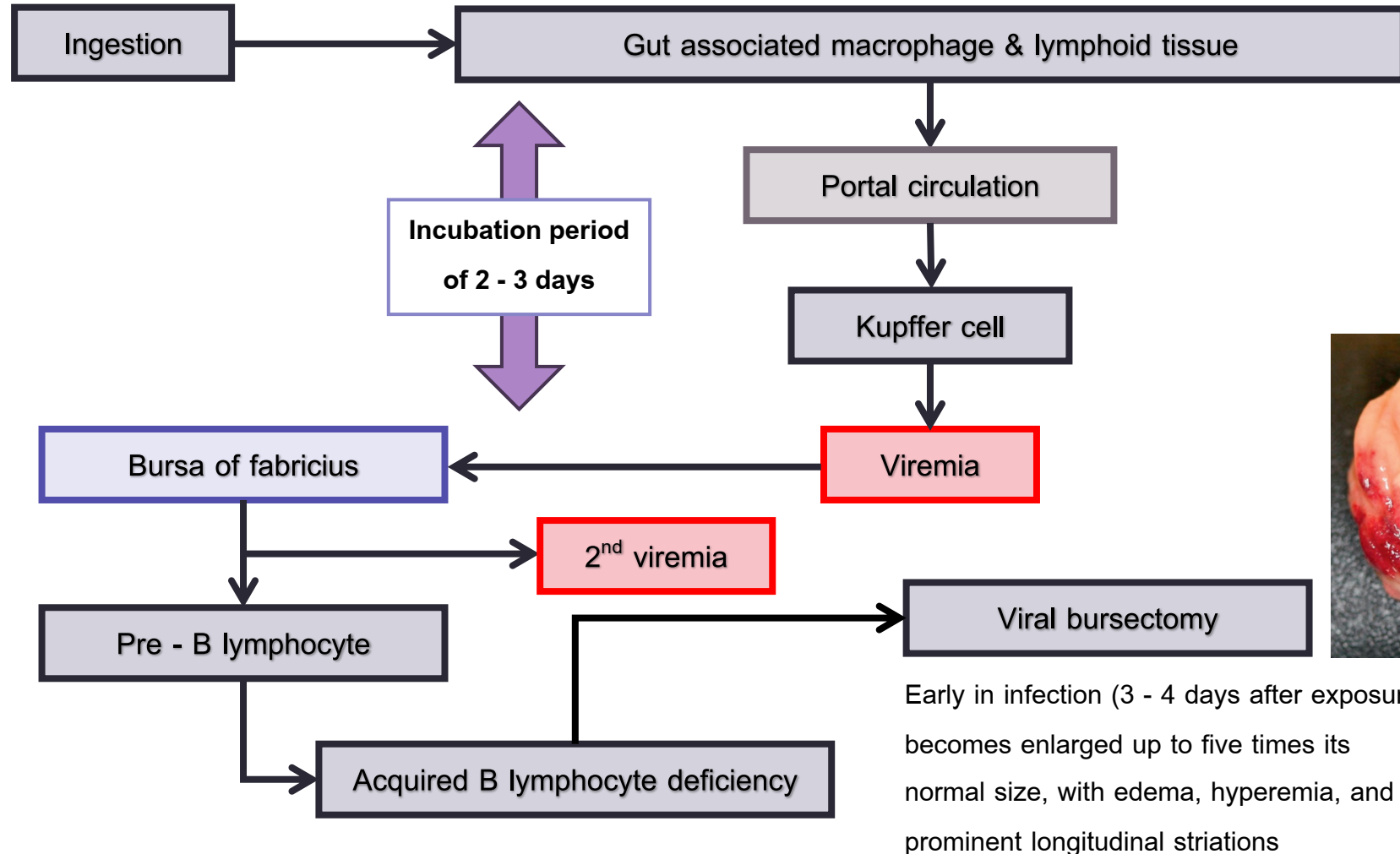
- **Baltimore classification**
 - Class III virus: double stranded RNA; dsRNA
- Replicate in the cytoplasm
 - Infectious bursal disease virus
 - replicates in both chicken and mammalian cells
 - Infectious pancreatic necrosis virus
 - replicates in fish cell lines incubated below 24°C
- Both avibirnaviruses and aquabirnaviruses can enter susceptible cells by the endocytic pathway

Infectious Bursal Disease Virus

- ชื่อพ้อง: Gumboro disease
- การก่อโรค: Chicken, turkey, duck
- อวัยวะเป้าหมายของ IBDV คือ pre-B lymphocytes ที่อยู่ใน Bursa of Fabricius
- แบ่งเป็น 2 serotypes (เฉพาะ serotype 1 ก่อโรครุนแรงในไก่ พบอัตราการตายถึง 50%)
- Virus excreted in the feces of infected birds for 2-14 days
- Route of transmission: contact and oral uptake



Transmission and Pathogenesis; IBD



Early in infection (3 - 4 days after exposure)
becomes enlarged up to five times its
normal size, with edema, hyperemia, and
prominent longitudinal striations

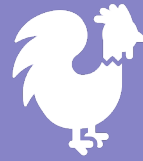
Clinical signs

- Symptoms: distress, depression, ruffled feathers, anorexia, diarrhea, trembling, and dehydration
 - Severe in chicks 3-6 weeks of age
 - Target organ: cloacal bursa
 - Chicks less than 3 weeks of age
 - may have subclinical infections because of their limited numbers of pre-B lymphocytes or from the presence of protective maternal antibodies
 - Birds older than 6 weeks
 - rarely develop signs of disease, although they produce antibodies to the virus
- Immunosuppression may persist in surviving birds
- Increasing susceptibility to other viral agents or bacteria

Diagnosis

- Immunofluorescence staining
- Gel diffusion test
- Electron microscopy
- Virus isolation
- Reverse transcriptase-polymerase chain reaction (RT-PCR) assay
- Neutralization assays
- Agar gel precipitin testing
- Enzyme immunoassay

Control and prevention



Stable and persists

More than 120 days in the farm environment
More than 50 days in feed, feces, and water



Biosecurity system

Phenolic-based compounds, iodine complexes, formalin, and chloramine compounds



Vaccination is the primary method of control



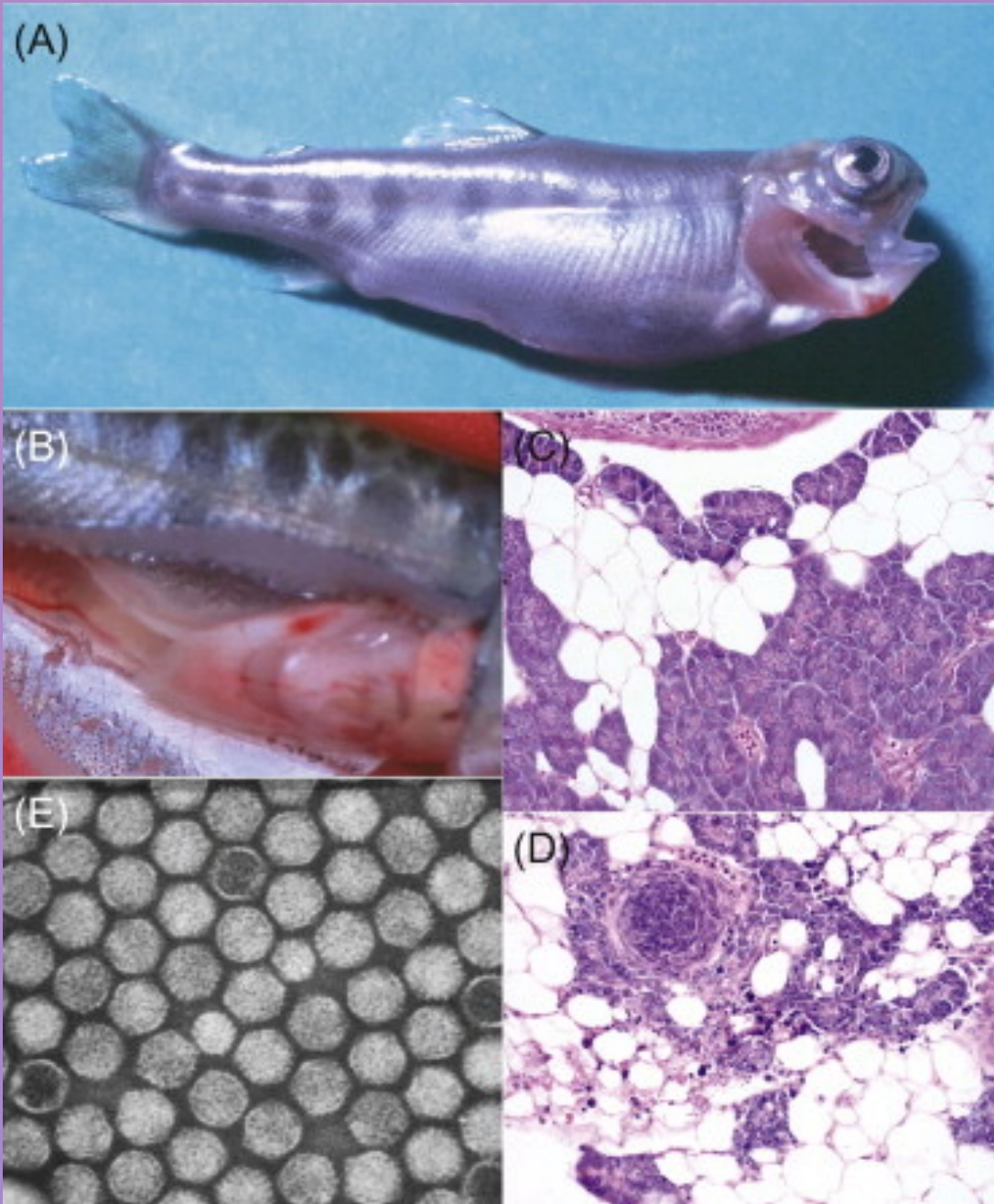
Infectious Pancreatic Necrosis Virus

- A highly contagious, lethal disease in several salmonid fish species
 - Fish older than 4 months often result in subclinical infections
 - Usually observed in trout fingerlings shortly after they commence to feed in freshwater
- พบไวรัสใน Atlantic salmon แต่ไม่พบการก่อโรค ปลาบางชนิดจะติดเชื้อแบบไม่แสดงอาการ เช่น carp, barbel
- Virus shedding from persistently infected carrier fish contributes to transmission of the virus to fish that cohabit the same waters
- The presence of the virus in eggs may result in vertical transmission to progeny
- The virus is highly stable under various environmental conditions, surviving for months in fresh or sea water



Clinical signs

- Fish are dark in color, swollen abdomen, mild to moderate bilateral exophthalmos, pale gills, cutaneous hemorrhages on the ventral body surface
- A frantic corkscrew swimming followed by periods of rest is commonly observed before death
- Mortality can range from 10-90%
- **Pathology**
 - Multiple petechiae may be present in visceral fat between intestinal ceca, especially in larger fish
 - Microscopic lesions: small to larger foci of necrosis in acinar cells of the pancreas



Diagnosis, control and prevention

Diagnosis

- Virus isolation
- Immunofluorescence
- Virus neutralization test
- ELISA
- RT-PCR

Control and prevention

- Control the hygiene, utilization of water sources free of fish (well water)
 - Disinfection of equipment with iodophors, culling of infected breeder fish and depopulation and sanitization if an outbreak occurs
- Multivalent vaccine



References

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